

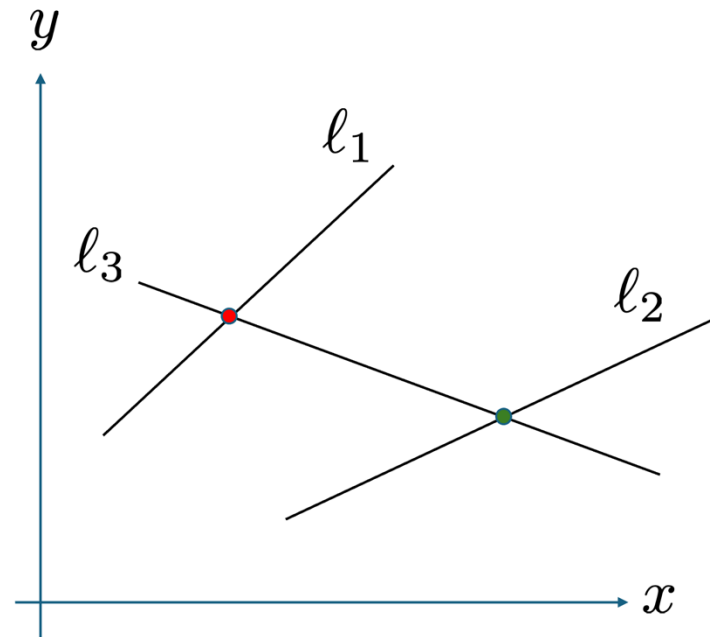


Coordenadas Homogéneas

(CAPÍTULO 2 – CV02A)

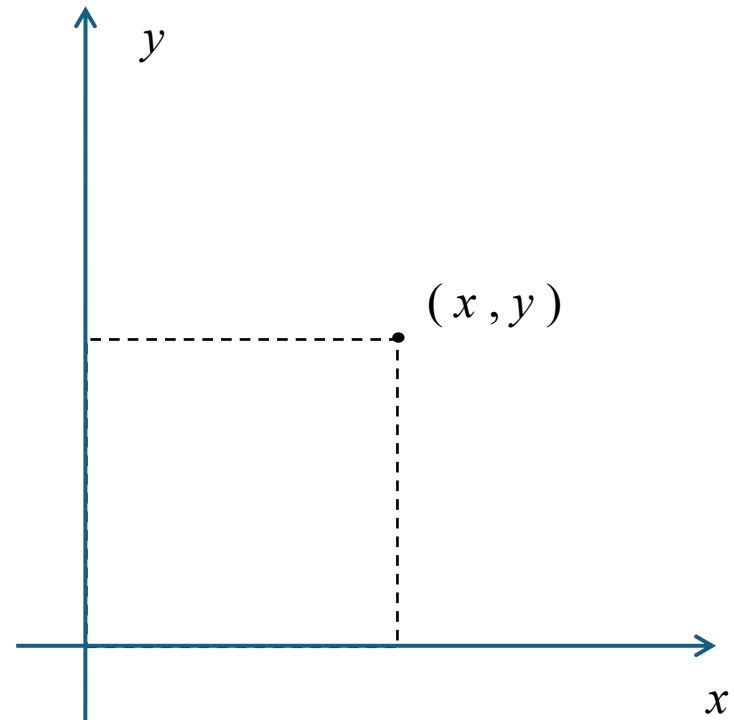
Domingo Mery 

Ciencia de la Computación
Universidad Católica – Chile
<http://domingomery.ing.uc.cl>



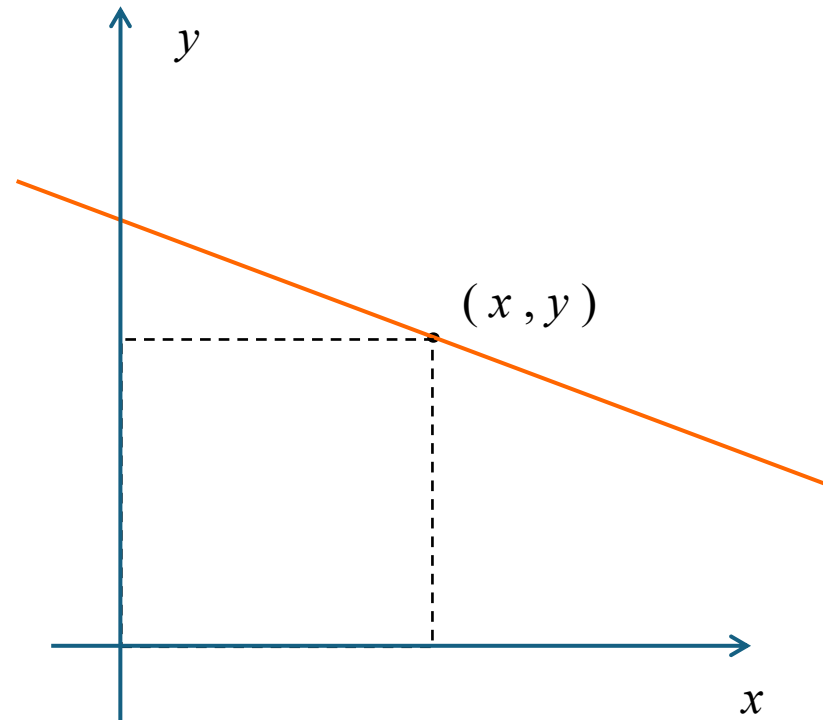
Puntos

[Point]

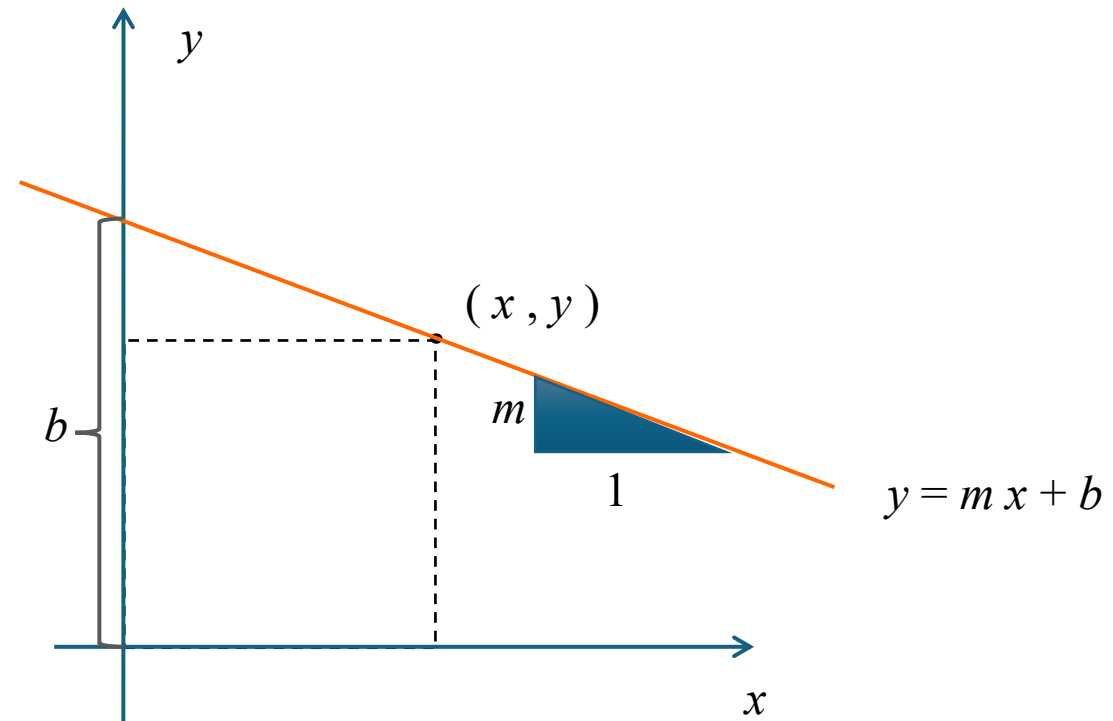


Líneas

[Line]



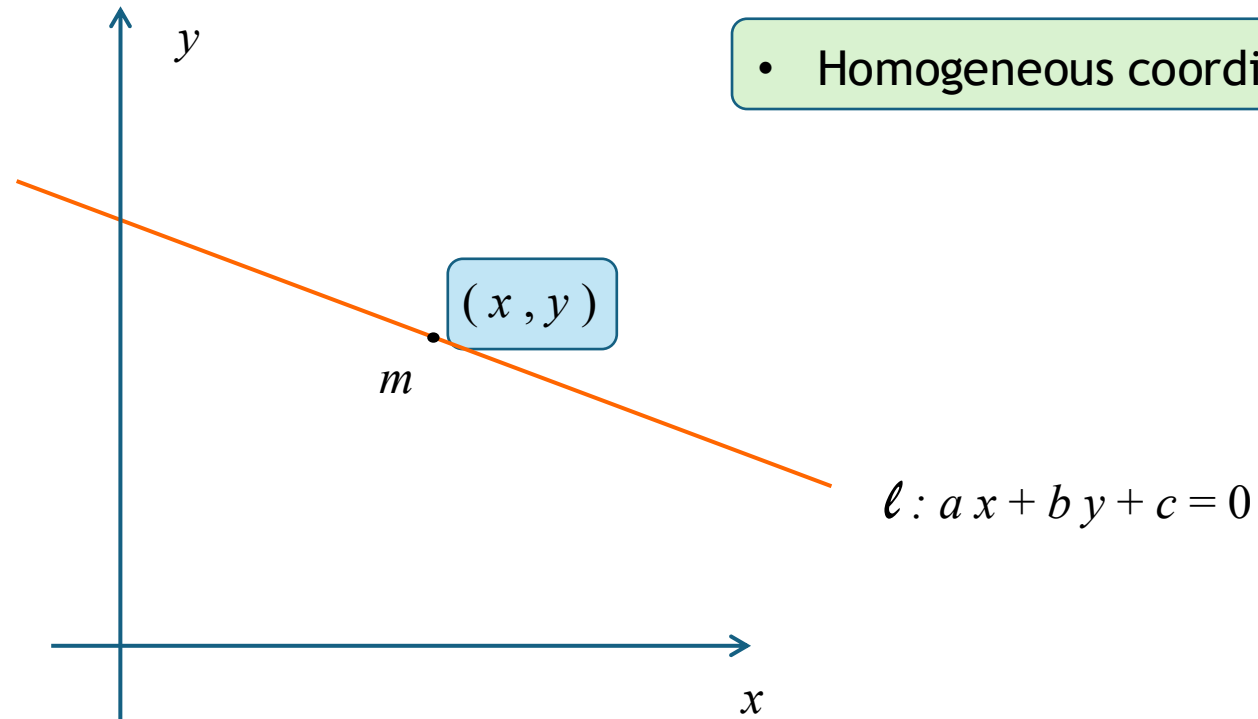
[Line]



[Line]

Two representations of m :

- Cartesian coordinates
- Homogeneous coordinates



$$\left. \begin{array}{l} \ell = [a \ b \ c]^T \\ \mathbf{m} = [x \ y \ 1]^T \end{array} \right\}$$

$$\begin{aligned} \ell \bullet \mathbf{m} &= 0 \\ \ell^T \mathbf{m} &= 0 \end{aligned}$$

Producto Punto

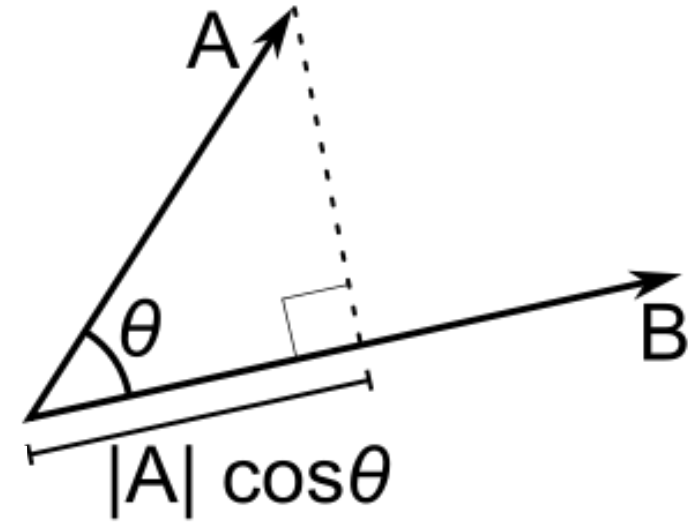
[Dot Product]

The dot product of two vectors $\mathbf{A} = [A_1, A_2, \dots, A_n]$ and $\mathbf{B} = [B_1, B_2, \dots, B_n]$ is defined as:

$$\mathbf{A} \cdot \mathbf{B} = \sum_{i=1}^n A_i B_i = A_1 B_1 + A_2 B_2 + \dots + A_n B_n$$

$$\mathbf{A} \cdot \mathbf{B} = \|\mathbf{A}\| \|\mathbf{B}\| \cos \theta,$$

where θ is the [angle](#) between $\mathbf{A}$ and $\mathbf{B}$.



[Dot Product]

The dot product of two vectors $\mathbf{A} = [A_1, A_2, \dots, A_n]$ and $\mathbf{B} = [B_1, B_2, \dots, B_n]$ is defined as:

$$\mathbf{A} \cdot \mathbf{B} = \sum_{i=1}^n A_i B_i = A_1 B_1 + A_2 B_2 + \dots + A_n B_n$$

$$\left. \begin{array}{l} \ell = [a \ b \ c]^T \\ \mathbf{m} = [x \ y \ 1]^T \end{array} \right\} \ell \bullet \mathbf{m} = a x + b y + c = 0$$

Producto Cruz

[Cross Product]

The cross product can also be expressed as

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$$

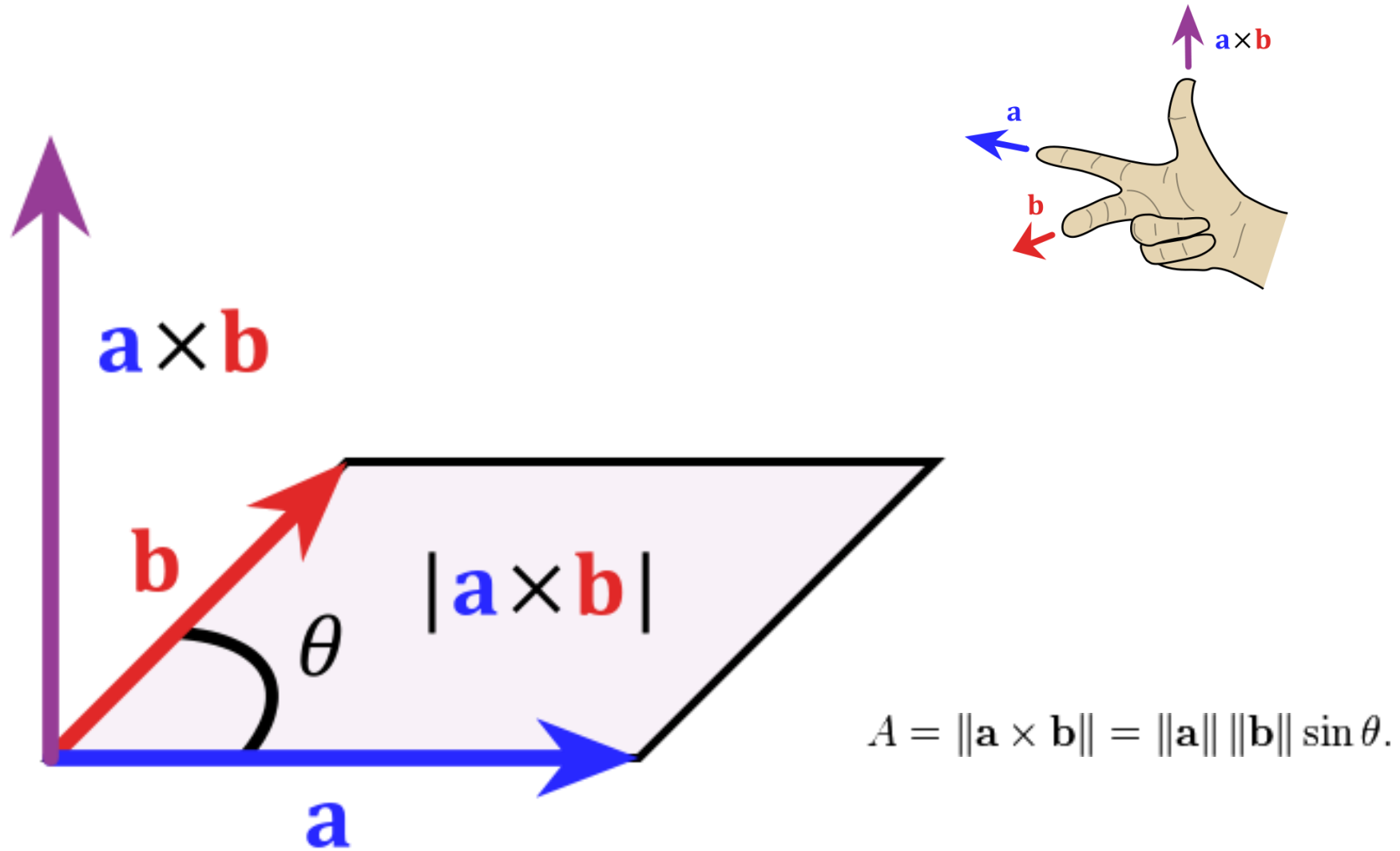
This determinant can be computed using

$$\mathbf{u} \times \mathbf{v} = (u_2v_3\mathbf{i} + u_3v_1\mathbf{j} + u_1v_2\mathbf{k}) - (u_3v_2\mathbf{i} + u_1v_3\mathbf{j} + u_2v_1\mathbf{k}).$$

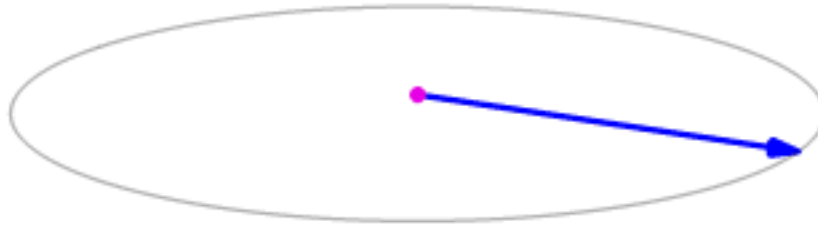
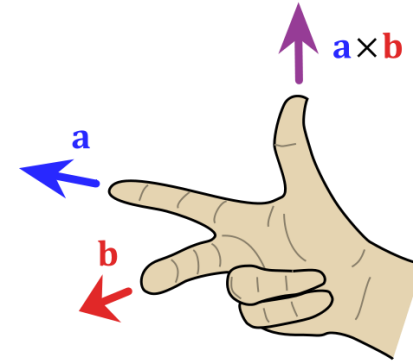
Using cofactor expansion along the first row instead, it expands to

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \mathbf{k}$$

[Cross Product]



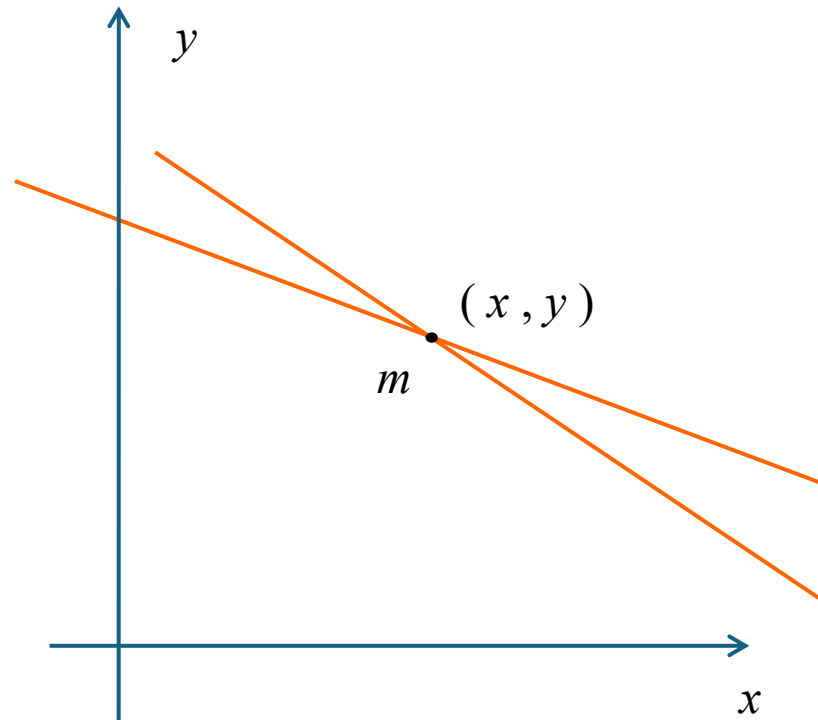
[Cross Product]



$$A = \|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \|\mathbf{b}\| \sin \theta.$$

Intersección de dos líneas

[Intersection of two lines]



We look for $\mathbf{m}$ so that

$$\ell_1 \bullet \mathbf{m} = 0 \quad \text{and} \quad \ell_2 \bullet \mathbf{m} = 0$$

$$\ell_1 : a_1 x + b_1 y + c_1 = 0$$

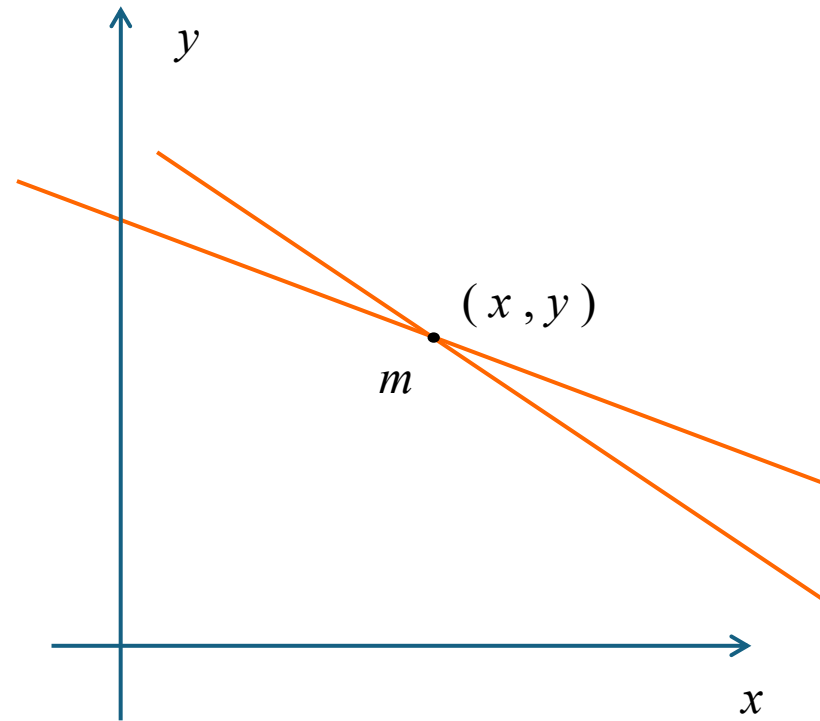
$$\ell_2 : a_2 x + b_2 y + c_2 = 0$$

$$\left. \begin{array}{l} \ell_1 = [a_1 \ b_1 \ c_1]^T \\ \ell_2 = [a_2 \ b_2 \ c_2]^T \end{array} \right\} \ell_1 \times \ell_2 = \mathbf{w}$$

$$\ell_1 \bullet \mathbf{w} = ?$$

$$\ell_2 \bullet \mathbf{w} = ?$$

[Intersection of two lines]



We look for $\mathbf{m}$ so that

$$\ell_1 \bullet \mathbf{m} = 0 \quad \text{and} \quad \ell_2 \bullet \mathbf{m} = 0$$

SOLUTION

$$\mathbf{m} = \ell_1 \times \ell_2$$

$$\ell_1 : a_1 x + b_1 y + c_1 = 0$$

$$\ell_2 : a_2 x + b_2 y + c_2 = 0$$

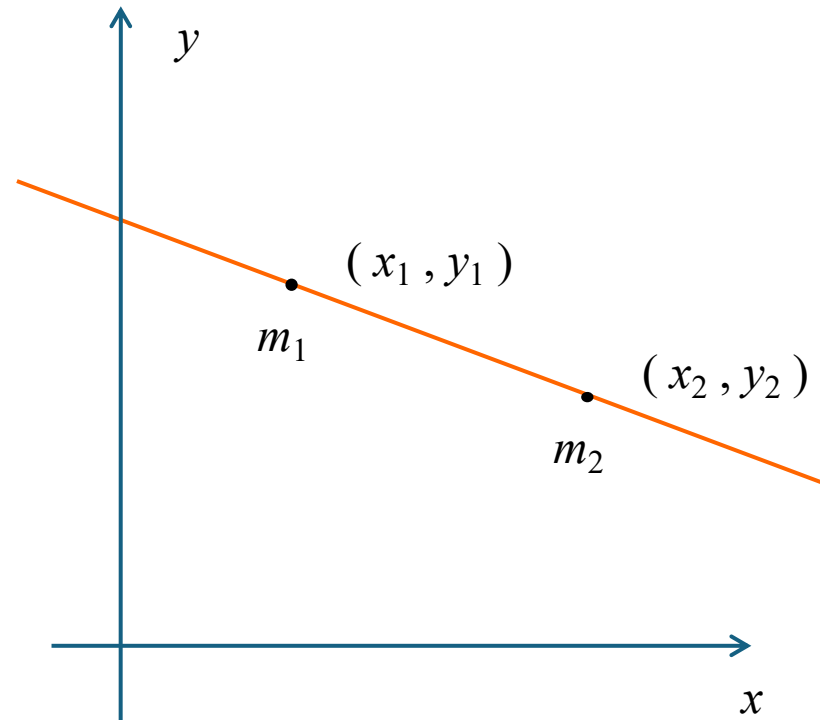
$$\left. \begin{array}{l} \ell_1 = [a_1 \ b_1 \ c_1]^T \\ \ell_2 = [a_2 \ b_2 \ c_2]^T \end{array} \right\} \ell_1 \times \ell_2 = \mathbf{w}$$

$$\ell_1 \bullet \mathbf{w} = 0$$

$$\ell_2 \bullet \mathbf{w} = 0$$

Línea que contiene dos puntos

[Line from two points]



We look for ℓ so that

$$\ell \bullet \mathbf{m}_1 = 0 \quad \text{and} \quad \ell \bullet \mathbf{m}_2 = 0$$

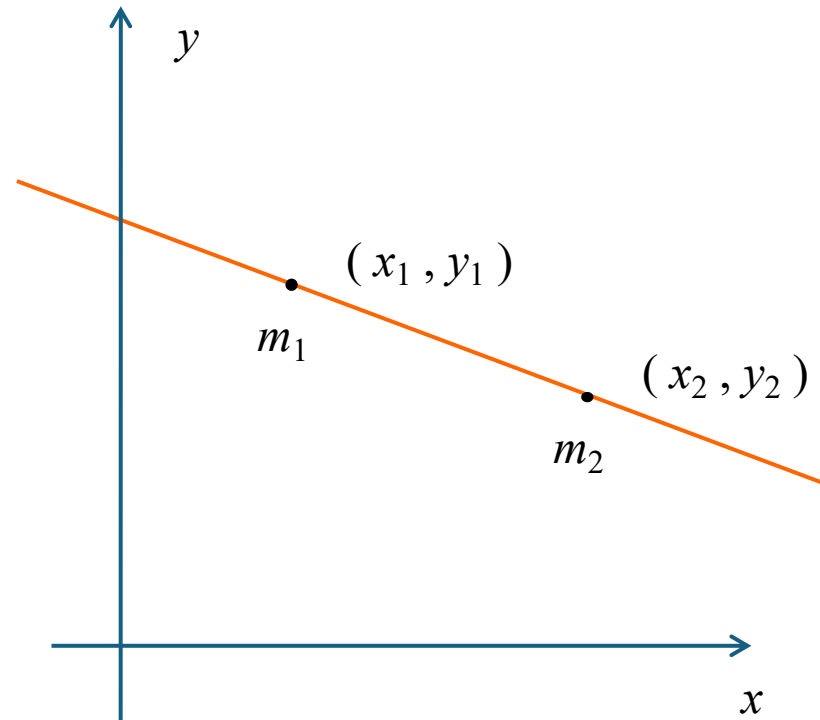
$$\ell: ax + by + c = 0$$

$$\left. \begin{array}{l} \mathbf{m}_1 = [x_1 \ y_1 \ 1]^T \\ \mathbf{m}_2 = [x_2 \ y_2 \ 1]^T \end{array} \right\} \mathbf{m}_1 \times \mathbf{m}_2 = \mathbf{w}$$

$$\mathbf{m}_1 \bullet \mathbf{w} = ?$$

$$\mathbf{m}_2 \bullet \mathbf{w} = ?$$

[Line from two points]



We look for ℓ so that

$$\ell \bullet \mathbf{m}_1 = 0 \quad \text{and} \quad \ell \bullet \mathbf{m}_2 = 0$$

SOLUTION

$$\ell = \mathbf{m}_1 \times \mathbf{m}_2$$

$$\ell: ax + by + c = 0$$

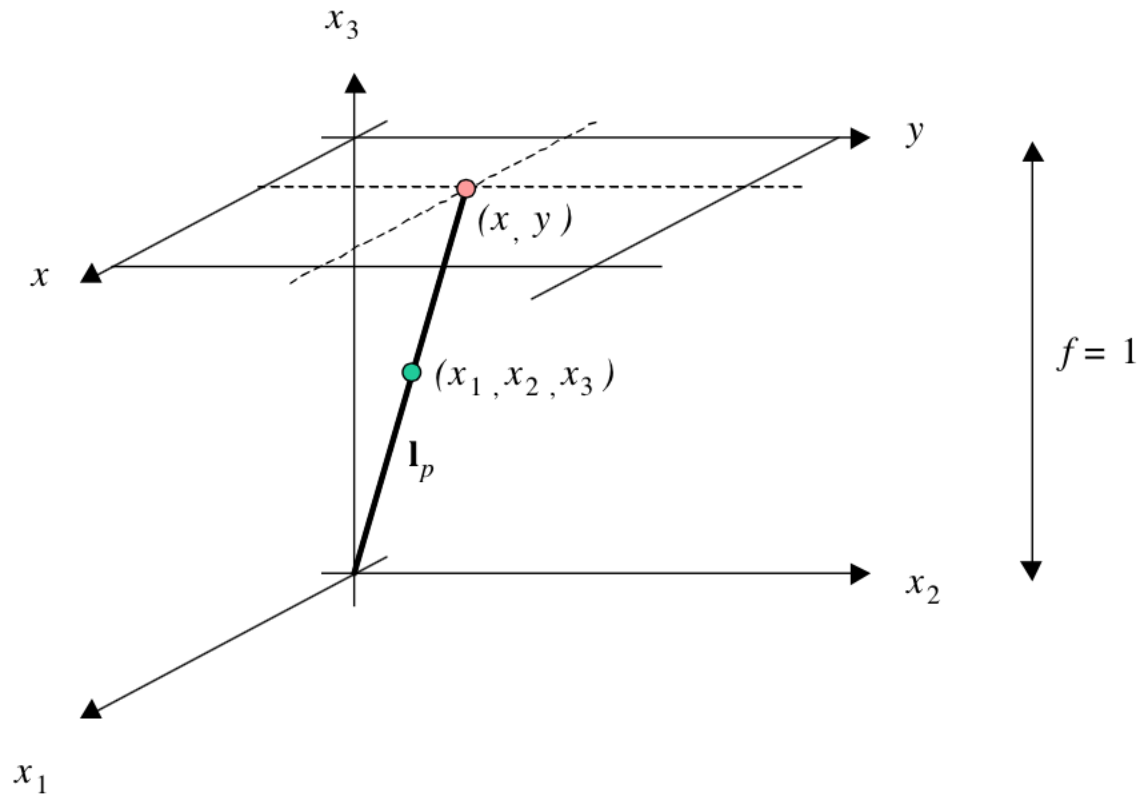
$$\left. \begin{array}{l} \mathbf{m}_1 = [x_1 \ y_1 \ 1]^T \\ \mathbf{m}_2 = [x_2 \ y_2 \ 1]^T \end{array} \right\} \mathbf{m}_1 \times \mathbf{m}_2 = \mathbf{w}$$

$$\mathbf{m}_1 \bullet \mathbf{w} = 0$$

$$\mathbf{m}_2 \bullet \mathbf{w} = 0$$

Coordenas Homogéneas

[Homogeneous Coordinates]



$$\mathbf{m} = [x_1 \ x_2 \ x_3]^T$$

Ejercicios

[Ejercicio 1]

Resuelva el siguiente sistema de ecuaciones usando la geometría proyectiva:

$$\begin{cases} x + 2y = 8 \\ x - 2y = 0 \end{cases}$$

$$\ell_1 = (1, 2, -8) \qquad \ell_1 \times \ell_2 = (-16, -8, -4)$$

$$\ell_2 = (1, -2, 0)$$

$$x = \frac{-16}{-4} = 4$$

$$y = \frac{-8}{-4} = 2$$

[Ejercicio 2]

Encuentre la recta que pasa por los puntos (10,20) y (20,30)

$$\mathbf{m}_1 = (10, 20, 1)$$

$$\mathbf{m}_2 = (20, 30, 1)$$

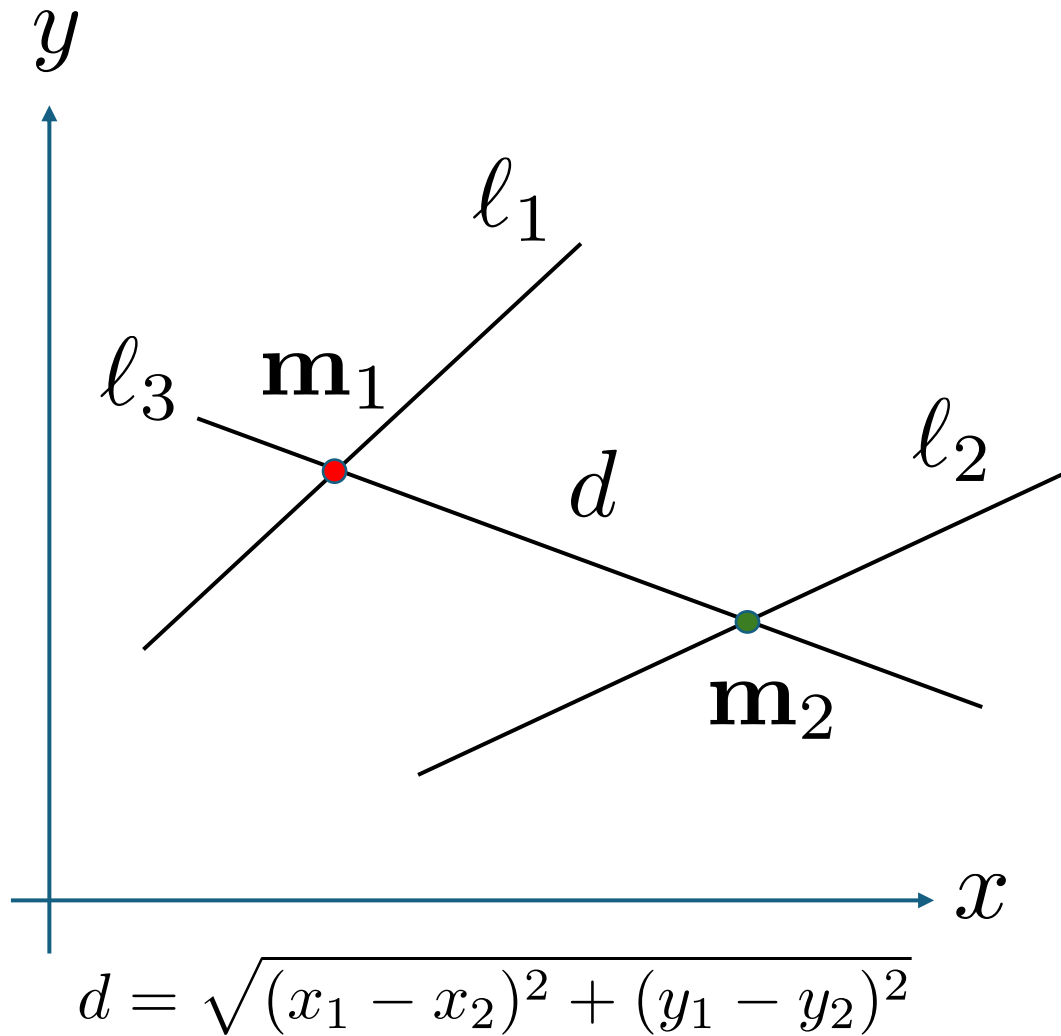
$$\mathbf{m}_1 \times \mathbf{m}_2 = (-10, 10, -100)$$

$$-10x + 10y - 100 = 0$$

$$y = x + 10$$

[Ejercicio 3]

Encuentre la distancia entre los puntos que intersectan las rectas



$$\mathbf{m}_1 = \ell_1 \times \ell_3$$

$$\mathbf{m}_1 = (a_1, b_1, c_1)$$

$$x_1 = \frac{a_1}{c_1}, y_1 = \frac{b_1}{c_1}$$

$$\mathbf{m}_2 = \ell_2 \times \ell_3$$

$$\mathbf{m}_2 = (a_2, b_2, c_2)$$

$$x_2 = \frac{a_2}{c_2}, y_2 = \frac{b_2}{c_2}$$